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Computer science has become most meaningful to me when projects on my laptop can directly impact people's lives. As I have progressed through my coursework and projects, I have become especially interested in cybersecurity, privacy, and networked systems because these areas shape whether the software people rely on is not only functional, but also trustworthy and safe. That interest has grown from a desire to use my skills to build technology that provides real value to real people. In developing my passion project, Homebound, I wanted to create something people could genuinely use and would actually benefit from. The app helps users create safety plans with automated check-ins and emergency contact notifications for overdue arrivals. Building Homebound made questions of security and privacy feel immediate rather than abstract, because an application centered on personal safety needs its users to be able to trust how their information is handled, how reliably the system works, and how thoughtfully it is designed.

That same motivation led me to value opportunities where I could pair what I was learning in class with real, tangible results. As a Student Research Assistant in the FuTURES^3 Lab at the University of Utah, I worked on improving OGHarn's Python harness generator by parallelizing it across CPU cores, reducing runtime by approximately 67% while achieving roughly 2x higher code coverage on target C libraries. Beyond the implementation itself, this experience taught me how to think more carefully about benchmarking, trade-offs, and systematic evaluation in systems work. Together, Homebound and my research experience have shown me that I am most motivated by problems at the intersection of security, reliability, and real-world impact, and they are the major reasons I want to pursue the blended master's program.

Another project that strengthened this idea for me has been my work on a haptic feedback system for a visually impaired marching band through the Empower Club at Cal Poly. In this project, I helped develop an ESP32 and ESP-NOW based wireless system designed to provide real-time haptic guidance to help musicians maintain formation and orientation during parades. What drew me to this project was not only the technical challenge but also the purpose behind it. Like Homebound, it gave me the opportunity to build something meant to help real people in a direct and meaningful way. At the same time, it deepened my interest in wireless systems and secure communication by showing how reliability, responsiveness, and thoughtful design become essential when technology is being used in dynamic, real-world environments. Working on a system where timing, coordination, and trust all matter reinforced my interest in building systems that people can depend on.

These experiences are a large part of why the blended master's program feels like the right next step for me. Through projects like Homebound and the haptic headband, along with my research experience, I have only reinforced my interests. I want to strengthen my foundation in cybersecurity, privacy, networked systems, and reliable software so I can approach these kinds of problems with greater rigor and a deeper understanding of how to design and evaluate them. The blended program would allow me to build on the momentum of my undergraduate coursework and projects while continuing to grow as a computer scientist focused on secure, dependable, and human-centered systems.

Cal Poly's blended master's program is especially appealing to me because it would allow me to continue building on this momentum in an environment that already aligns with how I learn best, with professors I have already created relationships with. One of the most valuable parts of my undergraduate experience has been Cal Poly's emphasis on Learn by Doing, in coursework, research, and independent projects. I want to continue learning in an environment where I am able to try to implement what I learned and where what I learn can directly support the kinds of systems I care about building. The blended program feels like a natural extension of the path I am already on, and would allow me to further my knowledge while staying connected to the Learn by Doing approach that has shaped me as a student.

I am still unsure of what exactly I want to do with my degree, but I am sure that I want to work on secure, reliable, and privacy-aware systems that everyday people can depend on. Especially in areas where software interacts closely with networks, devices, and real-world risk. I want to grow into an engineer who not only builds systems effectively, but also understands how to evaluate their trade-offs, anticipate vulnerabilities, and design them responsibly, as I have been learning how to do in my classes and with Homebound. Pursuing the blended master's program would help me strengthen that foundation and prepare me for more advanced work in cybersecurity and networked systems. It would allow me to continue developing the technical depth and research-minded approach I will need to contribute meaningfully to systems that have a real impact on people's lives.